



OPEN Publisher Correction: Subcellular visualization and quantification of cyanotoxin synthesis in cyanobacteria reveals distinct compartmentation

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Correction to: *Scientific Reports* <https://doi.org/10.1038/s41598-026-47303-1>, published online 19 April 2026

The original version of this Article contained an error in Figs. 1 and 4 where rendering caused blurring of the images.

The original Figs. 1 and 4 and accompanying legends appear below.

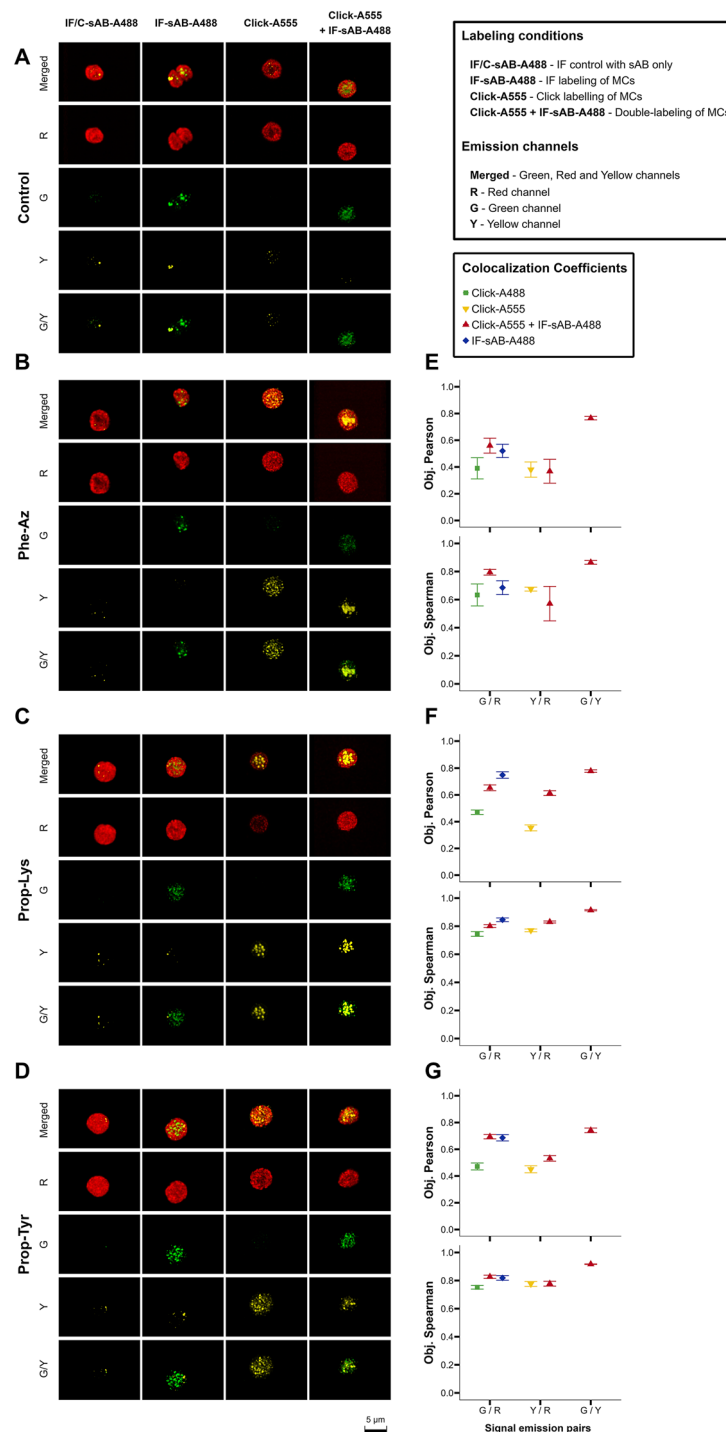


Fig. 1. Combining immunofluorescence (IF) labeling of MCs with chemoselective visualization of MCs in cells of *M. aeruginosa* strain Hofbauer (A–D). For this purpose, a double-labeling protocol for chemoselective labeling of clickable MCs (yellow fluorophore, click-A555-azide/alkyne) and the subsequent immunolabeling of MCs (green fluorophore, IF-sAB-A488) was established. *M. aeruginosa* cells were grown in the presence of non-AA, either Phe-Az, or Prop-Lys, or Prop-Tyr, to enable a click-chemistry reaction between clickable MC and A555-azide/alkyne. Control cells were grown in the absence of non-AA but treated identically. For immunofluorescence a monoclonal primary antibody (binding to the Adda side chain of the MC molecule) coupled to sAB-488 was used. The specificity of the secondary antibody sAB-488 was tested in absence of primary antibody (IF/C). Natural autofluorescence (AF) is observed in red. (E–G) The mean $\pm$ SE colocalization coefficients (Object Pearson and Object Spearman) were calculated to quantify the spatial correlation between different emission pairs: green channel (G) vs. red channel (R), i.e., IF-sAB-A488 vs. AF; yellow channel (Y) vs. R, i.e., click-A555 vs. AF; G/Y, i.e., IF-sAB-A488 vs. click-A555. For quantitative comparison of signal intensity between G and Y for both MC and another protein (RbcL, Rubisco large subunit) see Extended Data Fig. 1.

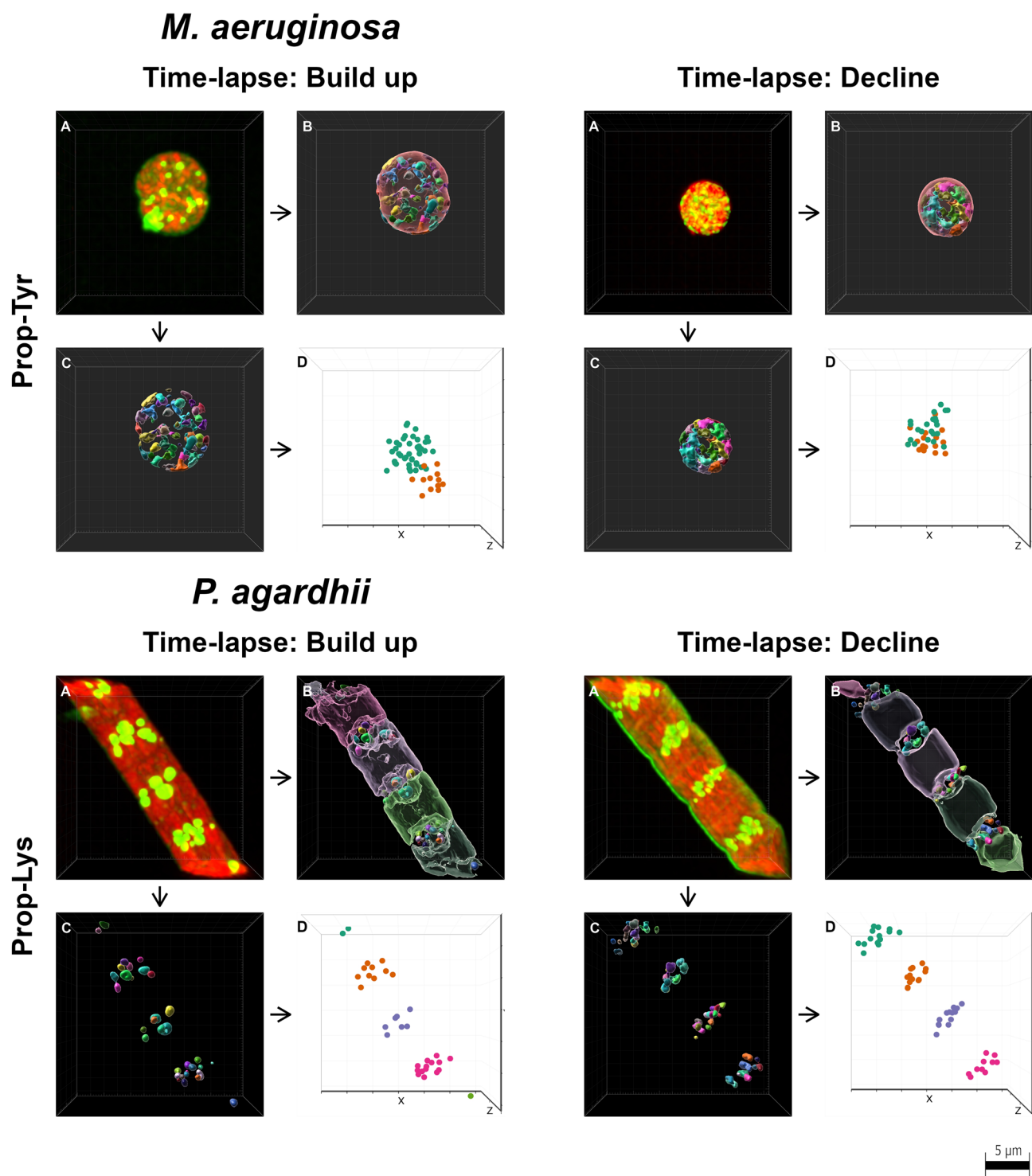


Fig. 4. (A) Workflow for processing high-resolution 3D microscopy images of *M. aeruginosa* strain Hofbauer grown in the presence of Prop-Tyr to produce clickable MCs and *P. agardhii* strain No371/1 grown in the presence of Prop-Lys to produce clickable APs. (B) Autofluorescence of cells (red signal) was used to estimate the cell shape via Imaris surface function (1) (C) The A488-click signal (in green) was used to calculate modeled entities (ME) using Imaris surface function (2) (D) Clustering analysis of the ME (x, y, z) coordinates was performed to quantify the intracellular ME distribution. The analogous workflows for other non-AAAs are shown in Extended Data Figs. 2 and 3.

The original Article has been corrected.

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